A Survey on Neural Radiance Fields and Gaussian Splatting in Satellite Remote Sensing

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***Abstract*—There has been a wide scope of satellite-captured images in the construction of 3D models of cities and landscapes. Still, traditional generic methods often struggle with various limitations, such as limited camera angles, changing sunlight, shadows, clouds, and the massive size of the satellite dataset. This survey demonstrates two modern techniques, Neural Radiance Fields and 3D Gaussian Splatting, that have proven effective for 3D Model Generation from satellite images. NeRF models work like a smart camera that learns various features from 2D images and reconstructs a 3D scene from those known features. This reconstructed 3D representation provides a prediction of how the architecture or the landscape would appear from a new angle. It is especially good at handling lighting changes and reflections. Conversely, 3DGS models build similar scenes using millions of tiny 3D particles, enabling faster and smoother training, and supporting real-time rendering. This survey discusses the challenges faced in urban and wide-scale reconstruction. This paper explains how these technologies are transforming five key areas of mapping. First, the generation of flat maps from complex 3D data. The implementation of large-scale mapping by slicing massive cities and historical sites into smaller, manageable chunks makes it easier for the computers to model them. These technologies are also capable of managing lighting and shadows by training the model how the shadows and lighting react to the appearance of an architecture to reveal the true shape and colour of the building. Additionally, they are highly accurate at filling the gaps when there is a shortage of satellite images. Finally, the survey covers monitoring structural changes, where the AI automatically detects new construction or architectural updates by comparing images taken across different seasons and weather conditions.**

***Index Terms*—Neural Radiance Fields, 3D Gaussian Splatting, Remote Sensing, 3D Reconstruction, Digital Twins, Neural Rendering, Urban Modeling**

TABLE I
LIST OF ABBREVIATIONS USED IN THE SURVEY

Abbreviation	Full Form
3DGS	3D Gaussian Splatting
BA	Bundle Adjustment
DSM	Digital Surface Model
EOGS	Earth Observation Gaussian Splatting
FPS	Frames Per Second
FSGS	Few-Shot Gaussian Splatting
GBM	Gaussian Building Mesh
GIS	Geographic Information Systems
LPIPS	Learned Perceptual Image Patch Similarity
MLP	Multilayer Perceptron
MSI	Multispectral Imagery
NeRF	Neural Radiance Fields
PAN	Panchromatic
PSNR	Peak Signal-to-Noise Ratio
RPC	Rational Polynomial Coefficient
SAR	Synthetic Aperture Radar
SfM	Structure-from-Motion
SSIM	Structural Similarity Index Measure
TDB	Top-Down Bias
TDOM	True Digital Orthophoto Map
TSDF	Truncated Signed Distance Function
UAV	Unmanned Aerial Vehicle
VRAM	Video Random Access Memory

I. INTRODUCTION

THE abundant amount of data captured by modern satellites has completely changed how we map the world [10]. We are no longer limited to flat representations; instead, the focus is on visualising physical environments exactly as they are [10]. With recent developments in deep learning and sensor hardware, building digital twin cities and complex 3D landscapes is now possible [1], [2]. These replicas allow decision-makers to analyse urban growth, detect environmental anomalies, and respond to emergencies with much greater precision [13]. To make this work, researchers have to extract precise 3D geometry from 2D image sets, creating a highly active research space that merges computer vision, photogrammetry, and remote sensing [2].

Recently, the development of Neural Radiance Fields (NeRF) has revolutionized the field of novel view synthesis and 3D modeling [3], [4]. NeRF uses a neural network to implicitly represent a scene as a continuous 5D function that maps spatial coordinates and viewing directions to volume density and radiance [3], [5]. This approach allows for the reconstruction of complex geometries and reflectance properties past simple color [5] In the remote sensing domain, NeRF has been adapted to handle the unique challenges of satellite imagery, such as Rational Polynomial Coefficient (RPC) camera models and strong non-correlation between images taken at different times due to shifting shadows and seasonal variations

[4]–[6] Remarkable extensions include S-NeRF and Sat-NeRF, which combines explicit shadow modeling and transient object filtering [5]–[7], and Ortho-NeRF, which produces True Digital Orthophoto Maps (TDOM) without requiring any prior 3D geometry [8], [9].

Even with the high-quality results produced by NeRF-based methods, they remain computationally challenging, usually requiring hours or even days for training and having to put up with slow rendering speeds [1], [9], [10]. To address these limitations, 3D Gaussian Splatting (3DGS) has recently surfaced as a better alternative [8], [11], [12]. Contrary to the implicit spatial representation of NeRF, 3DGS models a scene using millions of explicit 3D Gaussian ellipsoids [1], [10], [12]. This explicit representation enables real-time rendering (often exceeding 100 frames per second) and significantly faster training times, sometimes outperforming NeRF by a factor of 10 or more while maintaining or exceeding its visual quality [1], [7], [8].

A. *Scope and Organization*

The application of 3DGS to satellite remote sensing introduces a new edge in the field. Recent frameworks like EOGS and SatGS have extended 3DGS to satellite data by integrating appearance-adaptive strategies to account for large-scale seasonal color shifts and complex illumination changes [7], [10]. Additionally, 3DGS has shown superior performance in generating large-scale urban reconstructions and High-detail mesh reconstruction that can be seen in the PG-SAG and Gaussian Building Mesh (GBM) pipelines [11], [13]. Innovative hybrid methods like HyGS-TDOM have also been developed to mitigate reconstruction artifacts in sparse-view boundary areas of UAV and satellite strips by integrating few-shot learning techniques [14].

This survey aims to provide a thorough overview of the transition from implicit neural representations to explicit Gaussian-based models in satellite remote sensing. We look into the core methodologies of NeRF and 3DGS, their respective adaptations for large-scale earth observation, and the unique challenges posed by the satellite domain, which include geographical precision, multi-temporal appearance variations, and extreme perspective differences. By benchmarking the state of the art, this paper highlights the potential of these techniques to provide robust technical support for future urban digital twins and global-scale 3D mapping [1], [8]

II. Classification of Research Topics

To make sense of the many different ways researchers are using AI to map the Earth, we have organized the latest studies into five main categories. Traditional map-making often struggles with things like moving shadows, huge cities that crash computers, or having too few photos to work with. New AI technologies—specifically Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS)—are changing the game by **learning** how a scene looks from any angle. The following sections, detailed in Table II and Table III, break down these research areas: from creating perfect flat maps and massive 3D city models to fixing shadow problems, working with limited data, and tracking how buildings change over time.

III. Challenges and Research Questions in Remote Sensing

Solving long-standing challenges in 3D urban modelling has led to a switch from traditional photogrammetric workflows to neural rendering methods like NeRF and 3DGS. The literature reviewed in this survey addresses several questions that define the current state-of-the-art.

A. *Handling Multi-Temporal Radiometric Inconsistency*

Remote sensing datasets, particularly those that involve multi-view satellite constellations or multiday UAV missions, are in particular multi-temporal [7]. Particularly, images are obtained over vast time scales, leading to major changes in image appearance caused by changes in sun angle, atmospheric conditions, and phenology [7], [10]. As shown in Fig. 1, these changes can take various forms such as seasonal appearance changes, shadow changes, and object changes. These radiometric changes cause major noise in the neural supervision signal, challenging the multi-view consistency assumption essential for stable optimization [7], [11]. Therefore typical models tend to suffer from convergence instabilities where the model is unable to detect surface reflectance from lighting conditions or atmospheric changes [7], [10]. The major challenge in this context is how these neural models maintain geometric integrity when the underlyting supervision signal shows global and local radiometric changes due to seasonal or atmospheric changes [7].

To address geometric and radiometric consistency, neural frameworks make a distinction between static scene geometry from temporary appearance and lighting dynamics. Appearance-adaptive latent is employed to describe the style of acquisition, with SatGS using a hierarchical model in which global embeddings at the pre-image level describe broad atmospheric changes, while local embeddings at the pre-Gaussian level accommodate spectral fine-tuning [7].

To improve modelling of acquisition-specific sensor responses, for example, EOGS++ uses learnable affine transformations for each camera, which map the invariant 3D features to colour channels at a specified timestamp [10]. Instead of treating shadows as noise, S-NeRF and SatGS treat shadows as important information for geometry [7], [15]. This utilizes solar visibility fields and surrounding light from the sky to ensure consistency of shadows and buildings [7], [15]. Furthermore, in order to improve the robustness of the approach against temporary objects like moving vehicles, UPNeRF utilizes the uncertainty and visibility fields. This results in variance maps that reduce the loss for unreliable pixels, which prevents degradation of permanent scene structures [4].

B. *Solving Top-Down Bias in Vertical Reconstruction*

Remote sensing data is primarily captured from top-down. This results in **TDB**, where vertical structures like building

TABLE II
CLASSIFICATION OF RESEARCH TOPICS IN NEURAL AERIAL AND SATELLITE RECONSTRUCTION

Classification Topic	Brief Description	Representative Frameworks / Techniques
1. TDOM Generation	Traditional methods for creating **True Digital Orthophoto Maps (TDOMs)** use complicated techniques to figure out which parts of the scene should be visible in the final image. Because of this, the maps often contain problems like **blurry double images (ghosting) or distorted building shapes**. Newer approaches, such as **Ortho-NeRF**, fix many of these issues by using a more advanced rendering technique that removes perspective distortion. However, this method requires very high computing power and takes a long time to process. To get accurate results without needing so much computation, simpler and faster methods like **3D Gaussian Splatting (3DGS)** are used. These methods represent the scene in a more direct way, making map generation quicker and more practical. Fused framework like **HyGS-TDOM**, along with 3DGS with Few-Shot GS, including orthogonal splatting to yield highly accurate, full scene orthophotos [14].	• **Ortho-NeRF:** True-ortho volume rendering. • **Ortho-3DGS:** Depth-regulated optimization. • **HyGS-TDOM:** Hybrid 3DGS/FSGS.
2. Large-Scale Urban & Heritage	The generation of 3D models of huge cities or highly detailed historical architectures is in high demand. But the issue with this modelling is that it requires high computation power, and due to this, the computers might even run out of memory [1], [3]. The generation of 3D models of huge cities or highly detailed historical architectures is in high demand. But the issue with this modelling is that it requires high computational power, and due to this, the computers might even run out of memory. To solve this, researchers use **divide-and-conquer** strategies [11]. Advanced AI frameworks break the scene down into smaller chunks, instead of processing the entire city at once [8], [11]. For example, methods like PG-SAG group elements by automatically recognizing and separating building shapes, allowing computers to process the environments piece by piece [11]. Additionally, the experts compare these AI tools with 3D photo-matching methods to digitally preserve the museum collections and heritage buildings [17].	• **PG-SAG:** Semantic grouping. • **Block-NeRF:** Scene partitioning. • **Heritage3DMtl:** LOD3 models.
3. Shadow & Illumination	While generating a 3D map from the satellite images from over years or months, a major issue with this generation is that the scene changes consistently. The sun's angle shift causes the shadows to move around the object, and also the temporary objects, like cars, appear and disappear within the images [7]. This makes it confusing for the 3D mapping software. To fix this issue, new smart models like S-NeRF and SatGS were developed, which did not just focus on memorizing the pictures, but also calculated the sun's position to separate the true shape of the building from its temporary shadow [6], [7]. Some of the well-designed tools can even adapt to the visual changes in seasonal variations like winter snow, and automatically ignore moving cars so they do not distort the map [7]. To train these AI models, researchers created a massive S-EO dataset, which provided thousands of examples of perfectly mapped shadows that act as a training guide to help computers build flawless, shadow-free 3D digital twins of the Earth [19].	• **S-NeRF / Sat-NeRF:** Solar visibility modeling. • **SatGS:** Appearance-adaptive 3DGS. • **EO-NeRF:** Shadow supervision.

TABLE III
CLASSIFICATION OF RESEARCH TOPICS IN NEURAL AERIAL AND SATELLITE RECONSTRUCTION

Classification Topic	Brief Description	Representative Frameworks / Techniques
4. Few-Shot Synthesis	When drones capture photos to create 3D maps, they only capture a few photos of the extreme edges along their flight paths. Standard 3D models struggle with this sparse data and often get confused, resulting in the creation of distorted 3D models and messy shapes because they simply lack sufficient visual information [4], [14]. To resolve this issue, researchers are developing few-shot techniques that are capable of building 3D scenes from limited images [14]. Latest tools like UP-NeRF use an uncertainty-guided approach to identify which parts of an image the computer is unsure about, learning the clear, easy parts first before tackling the complex details [4]. Other methods, like Few-Shot Gaussian Splatting (FSGS), use smart depth-guessing tricks, which helps the computer understand structural distances, ensuring the resulting 3D map to stay sharp and accurate even when visual data is highly limited [14].	• **UPNeRF:** Uncertainty-guided. • **FSGS:** Proximity-guided unpooling. • **DS-NeRF:** Depth supervision.
5. 3D Change Detection	Figuring out what has changed in the 3D environment over time, such as a newly constructed building or a removed tree, is highly challenging, especially when the lighting or weather drastically changes between photographs. Traditional mapping software easily gets confused with these environmental differences and usually requires pictures that are taken from the exact same spot. To solve this issue, a 4D modelling technique, such as a framework called Gaussian Difference, is being developed. Instead of comparing flat 2D pictures, this tool builds a 3D digital model of the area both before and after the change occurs. It automatically acts like a digital supervisor, assigning tracking IDs to every individual object and comparing them across time. This allows the system to easily pinpoint true structural changes from any viewing angle, completely ignoring misleading differences caused by shifting shadows or varying sunlight [20].	• **Gaussian Difference:** 4D GS. • **C-NERF:** Directional consistency.

front-view suffer from significant information loss and structural ambiguity. This large difference in perspective makes the establishment of geometric correspondences between aerial rooftops and facades on the ground level an inherently ill-posed problem [12]. Without the availability of sufficient oblique imagery to provide side-view context, neural models often have difficulty disentangling the conflated foreground buildings from the background terrain, leading to **jagged** or **sawtooth** effects that refers to geometric misshaping and serrated irregularities that appear along the edges, boundaries, and bases of reconstructed vertical structures, such as building facades or a complete absence of the vertical elements. As shown in Fig 2, the key challenge that how neural rendering frameworks can solve the TBD inherent in vertical remote sensing imagery to accurately reconstruct vertical building surfaces and eliminate geometric features like jagged edges or the merged surface with ground.

Recent literature shows that this problem is increasingly addressed by neural frameworks leverage explicit geometric control of 3DGS and specialized loss functions to recover high- fidelity vertical structures. To address the **TDB** limitation in vertical reconstruction, these modern frameworks combine oblique and multi-perspective imagery for providing the necessary sideview context that is missing in predominantly nadir (top-down view) datasets [2], [5].For example, pipelines like Gaussian Building Mesh (GBM) and Ortho-3DGS utilize the benefits of multiple angular views to recover high-fidelity vertical geometry that is hidden or deformed in standard satellite-based reconstruction [8], [13]. This is further improved in cross-view tasks through the use of pseudo-aerial views at a significant pitch angle, like 45° or 60°, that allow the model to penetrate foliage and capture building sides shared by both ground and aerial views [12]. To complement this data integration, PG-SAG uses semantic- aware scene partitioning to address structural uncertainty between foreground buildings and surrounding background [11]. By using subgroups for scene partitioning, the model ensures that vertical facades do not overlap with the terrain [11]. PG-SAG introduces a

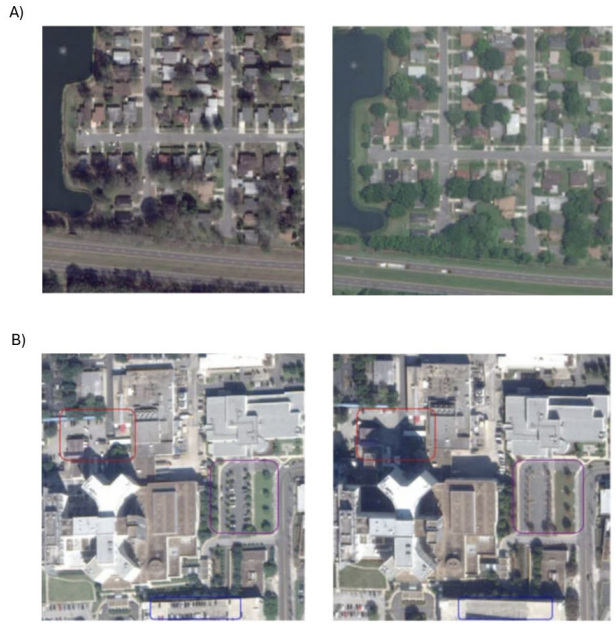

Fig. 1. Multi-temporal radiometric inconsistency in remote sensing imagery: A) seasonal appearance variation and B) shadow geometry change and appearance of transient objects, such as vehicles, which impact photometric consistency and make neural rendering optimization difficult [7].

boundary- aware normal loss to eliminate "jagged" or "saw-tooth" effects commonly seen in traditional methods. This loss term reformulates the optimization process along the detected edges in a way that reduces normal vector ambiguity, which ensures better structural preservation [11]. Lastly, GBM-like frameworks provide better vertical fidelity by using morphological operations and/or contour simplification on building masks. This makes sure that the building bases are well-defined, which reduces any errors resulting from having no building masks fully cover the ground and the vertical facades of the building.

C. Massive Data Size

There are significant hardware challenges in the transition from object-centric representations to city-scale digital twin representations due to synchronous optimization of millions of Gaussian primitives over entire urban districts, which over-reaches traditional hardware's VRAM limit, and this results in prohibitive computational latency [11]. For example, urban scenes at large scales require a large number of Gaussian functions to capture complex details, which can overwhelm GPU memory during both training and real-time rendering [8], [11]. As shown in Fig. 3, the major challenge in this is how explicit neural representations be scaled to city-scale digital twins without going beyond hardware limitations for VRM or compromising the high rendering speeds required for interactive applications.

To address these challenges due to massive data scale, the current literature explores divide-and-conquer architectural approach based on spatial and semantic partitioning [9], [11]. One of the major direction is spatial tiling, which is explored in Ortho-NeRF framework, where large urban scenes are divided into geographically aligned tiles that can be reconstructed independently and later stitched into a coherent True Digital Orthophoto Map (TDOM) [5], [9]. This reduces the memory load of monolithic optimization and helps to avoid GPU memory overflow [9], [11]. Recent methods like PG-SAG extend this idea through semantic-aware grouping, where building regions are organized according to visibility relationships across registered images to allow more parallel and memory-efficient Gaussian optimization [11], [14]. This is mainly useful for preserving fine architectural details without depending heavily on image downsampling [11]. Simultaneously, other works explore parameter-efficient representations, that includes adaptive resolution control and multi-resolution hash encoding (as used in Instant-NGP), to reduce memory consumption while maintaining rendering quality [2], [5], [8], [9]. These methods aim to assign model capacity more selectively, rather than distribute uniformly across the entire scene [9]. Finally, the literature explores progressive chunking as a solution for seamless integration. These strategies investigate how aligning chunk boundaries through camera overlap can reduce the inconsistencies and artifacts that typically occur during the final merging of large-scale scene segments [8].

D. Mixing Cameras: Multi-Sensor and Cross-Model Fusion

Integrating data from multiple satellite platforms is technically demanding because of significant variations in spatial resolutions and sensor characteristics over different missions [10]. When integrating datasets, it requires complex alignment and radiometric normalization, which can introduce noise and reduce structural accuracy. These limitations become especially important when spectral content from several sensors needs to be combined with high-resolution geometric data. As shown in Fig 4, the main challenge here is how can high-resolution structural information be effectively merged with spectral data from diverse sensors without incurring artifacts and inconsistencies related to traditional preprocessing steps, such as pansharpening [10].

As indicated by literature, particularly EOGS++ framework that proposes a shift toward pipelines with **virtually zero** required preprocessing by operating directly on raw data [10].

To avoid spectral artifacts and spatial blurring that is common in external pansharpening, EOGS++ introduces a "Direct Rendering" approach. This approach works directly with raw high-resolution panchromatic (PAN) images, without the need for low-resolution multispectral data for geometry estimation, to maintain the maximum amount of structural details. This is done by creating a three-channel representation by duplicating single-channel PAN data, and the model directly optimizes a stable 3D core against high-resolution observations [10].

Any misalignment caused due to the use of different camera types and timestamps is corrected using internal bundle adjustment (BA) embedded directly into the training loop. The framework calculates displacement fields to align rendered

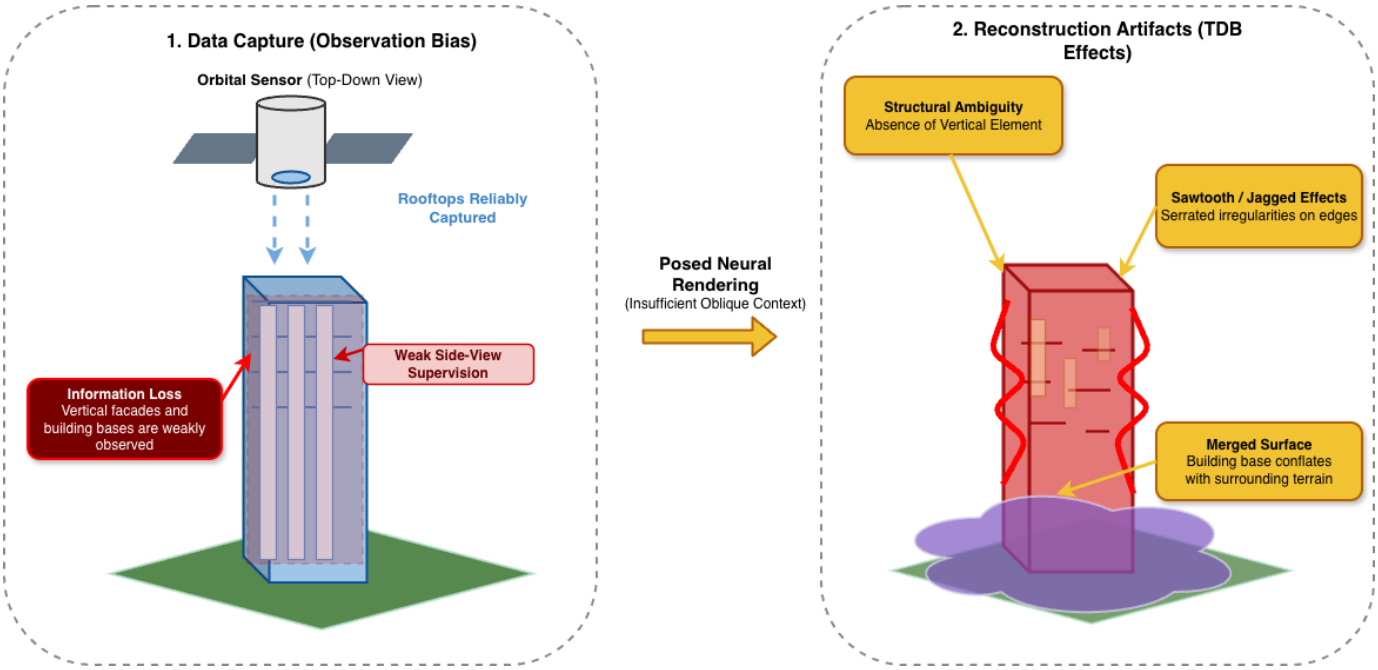

Fig. 2. The effect of Top-Down Bias (TDB) on vertical reconstruction. Cleanly top-down data capture (Left) results in extreme information loss on building facades. This lack of side- view context causes major 3D reconstruction artifacts (Right), including structural ambiguity, **jagged** or **sawtooth** edges, and the merging of the building base with the surrounding terrain.

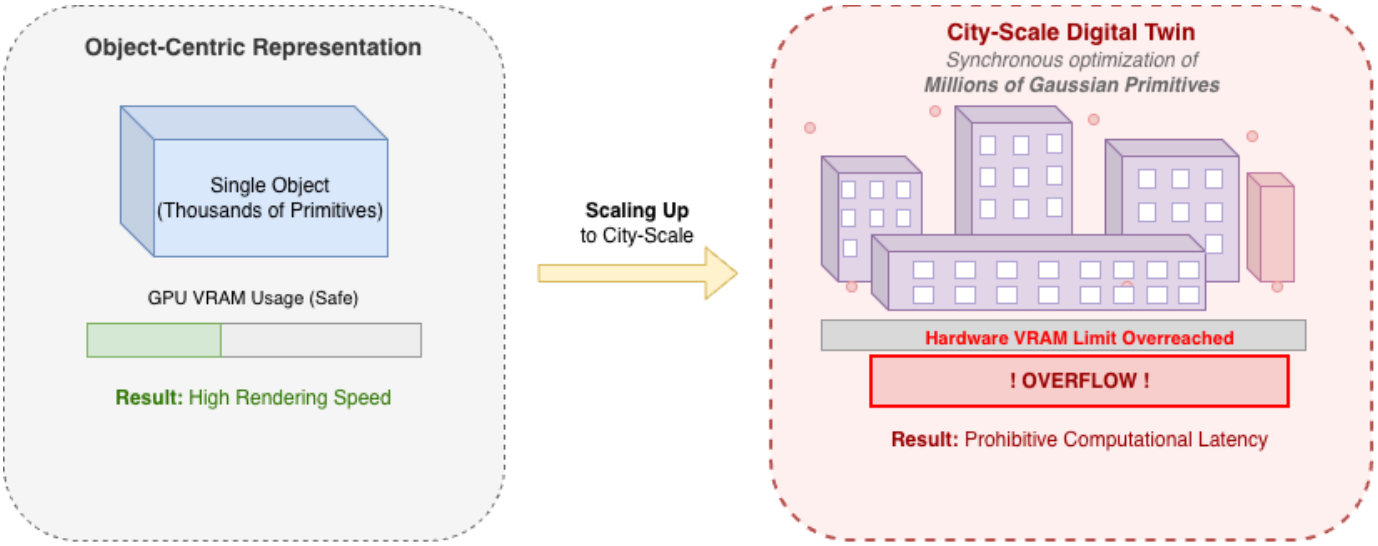

Fig. 3. An example of the scalability problem in explicit neural representations: Object- centric scenes can be reconstructed with high rendering speed and manageable GPU memory usage, but scaling city-scale digital twins requires synchronous optimization of millions of Gaussian primitives, which can introduce prohibitive computational latency and exceed hardware VRAM limits.

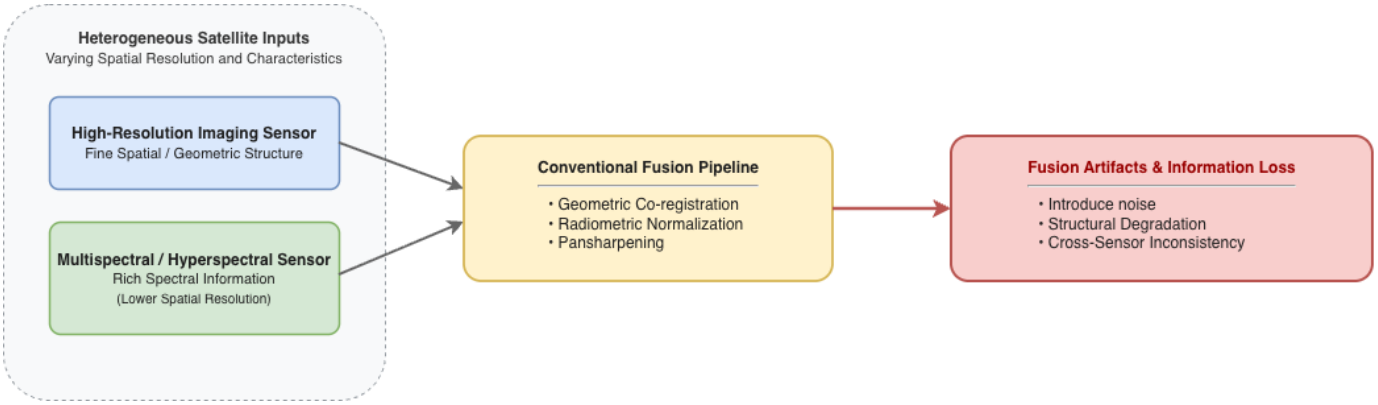

Fig. 4. The difficulties in cross-model and multi-sensor satellite fusion. Before fusion, co- registration, radiometric normalization, and pansharpening are required for lower-resolution spectral data from heterogeneous sensors and high-resolution structural imagery. These preprocessing procedures usually result in cross-sensor inconsistencies, noise, and structural degradation.

views with reference images using readily available optical flow estimators like RAFT. This eliminates the need for external optimization software and corrects pre-camera pixel offsets and pose errors. This integrated approach ensures that structural data from different sensors is geometrically consistent and produces much sharper results than the initializations using raw RPC data [10].

IV. Fundamentals of Neural Rendering

A. Neural Radiance Fields (NeRF)

Neural Radiation Fields (NeRF) are amongst the most prominent techniques that utilise continuous implicit neural representations for 3D scene reconstruction rather than explicit geometries like meshes and point clouds. This technique has outperformed others in the creation of photorealistic 3D models from 2D images by encoding complex scene geometry and appearance directly into the neural network's weights [14]. At the core, NeRF utilises a fully connected deep network, typically a Multilayer Perceptron (MLP), to encode these scenes into network weights [8].

As mentioned in Fig. 5, the reconstruction process models the scene as a continuous 5D function. The MLP takes a 5D coordinate as input—comprising a 3D spatial location (x,y,z) and a 2D viewing direction (θ, ϕ) — and outputs the volume density and view-dependent emitted radiance (RGB colour) at that specific spatial point. The NeRF's framework works with a differentiable volume rendering pipeline as its core. This works by casting the rays into the scene and sampling spatial points. The system combines the radiance and opacity using alpha compositing. This enables the network to learn a highly accurate 3D representation from a set of 2D images by reducing the discrepancy between rendered and ground-truth views [3]. The network is optimised end-to-end by minimising the photometric loss between the rendered synthetic images and the actual ground-truth multi-view training images [6].

In the domain of 3D urban reconstruction, NeRF models demonstrate strong capabilities for constructing 3D geometry from appearance, such as reflections and translucency, which

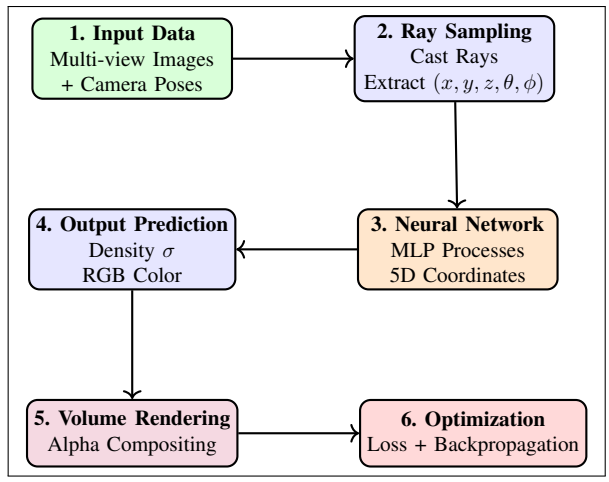

Fig. 5. Workflow of the Neural Radiance Fields (NeRF) pipeline.

traditionally confuse standard Structure-from-Motion (SfM) photogrammetry. Consequently, NeRF has been effectively adopted for digitization of cultural heritage for complex artifacts [3]. Furthermore, NeRF's use has also been proven helpful in large-scale urban modelling and remote sensing [7]. Advanced frameworks like Ortho-NeRF generate a True Digital Orthophoto Map (TDOM) by the utilization of orthogonal ray sampling instead of central projection. This overrides the problems of geometric distortions and jagged edges [8], [9]. Similarly, the S-NeRF (Neural Radiation Fields for Street View) models the ambient sky lighting and illumination to reconstruct precise topography even in the case of changing shadow conditions [6], [7].

Despite having many remarkable successes, the standard NeRF model faces significant limitations. The model relies on a number of MLP evaluations per ray, which introduces a computational bottleneck. This results in intensive training cycles - often hours or even days - and the rendering speed is far too low for real-time applications [3], [9]. Additionally, NeRF requires dense multi-view supervision and also struggles

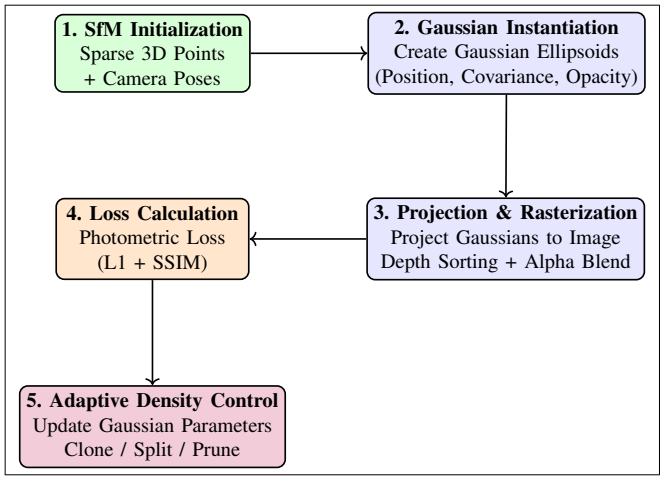

Fig. 6. Workflow of the 3D Gaussian Splatting (3DGS) pipeline. The process begins with SfM initialization, followed by Gaussian instantiation, rasterization-based rendering, photometric loss evaluation, and adaptive density control for scene refinement.

with overfitting when provided with sparse observational data, which is considered to be a common issue in aerial photogrammetry [4].

B. 3D Gaussian Splatting (3DGS)

3D Gaussian Splatting (3DGS) has rapidly emerged as one of the transformative methods for high-fidelity 3D scene reconstruction and novel view synthesis. This provides a highly effective and better alternative to the previous implicit neural representation, like NeRF [8], [11], [14]. Unlike the NeRF, which relies on multi-layer perceptrons (MLPs) and highly computationally expensive volume rendering, the 3DGS utilizes explicit scene representation using millions of tiny 3D Gaussian Ellipsoids [8], [14]. Each of the Gaussian Primitives is parameterized by some of its key parameters: 3D centre position, covariance (which defines anisotropic shape and scale within the 3D space), opacity, and color information, which is represented by the spherical harmonics [1], [10], [14].

As shown in Fig. 6, the reconstruction pipeline generally initializes the Gaussian Primitives using a sparse point cloud, which is generated via Structure-from-Motion (SfM) [1], [14]. During the optimization phase, 3DGS projects these Gaussians onto a 2D plane using highly parallelized, tile-based rasterizers (Rasterizer is a component in the graphics pipeline that converts 3D scenes data (such as points, polygons, or primitives) into 2D pixels on the screen for rendering an image). These projected 2D Gaussians are depth-sorted and combined together using alpha blending to synthesize the final image. The network then adjusts the Gaussian parameters to reduce the photometric loss between the rendered output and the ground-truth images by controlling density via splitting, cloning, or pruning Gaussians. This explicit formulation overrides the empty space calculations, enabling exceptionally faster training and real-time rendering (exceeding 100 frames per second) [8], [14].

In the field of applied 3D reconstruction, 3DGS has showcased versatile performance across various domains. For the large-scale urban modeling, the researchers address the GPU memory constraint by dividing the scenes into smaller chunks/regions, which allows them to process every sub-region in parallel [11]. Additionally, 3DGS has also improved in True Digital Orthophoto Maps (TDOMs) generation by replacing perspective camera models with orthogonal projections, which directly produce distortion-free mapping results. To address geometric degradation in sparse-view scenarios, such as the UAV flight boundaries, some frameworks use common learning methods like few-shot learning (FSGS) and molecular depth supervision to better constrain and preserve the scene geometry. Additionally, specialized variants for satellite remote sensing, such as SatGS and EOGS++, utilize appearance adaptive adjustments and camera refinement to better handle seasonal change, temporary objects, and shadows [7], [10].

V. Neural Radiance Fields (NeRF) in Earth Observation

A. Illumination and Shadow Modeling

In a multi-view satellite photogrammetry, images of the same area are often taken at different times of the day and across different seasons. This changes causes noticeable changes in appearance, most important it also capture the changing sun angles and temporary objects like vehicles and pedestrians [7]. The standard NeRF models often struggle with these uncontrollable variations, which often results in creation of visual artifacts, containing blurry textures or inaccurate geometry, specifically in the shadowed regions [15].

To address this challenges, Shadow Neural Radiance Fields (S-NeRF) was introduced which was an explicit shading model which was specifically designed for Earth Observation [16]. The S-NeRF separates the lighting from the scene from its inherent surface properties by predicting the base color (albedo) along with the effect of direct sunlight, ambient skylight or climatic vision [15], [16]. By the utilization of the known solar direction vector for every input image, the network predicts an solar visibility function (which performs as a structural shadow mask) and a diffused sky color [15].

This explicit separation supports the S-NeRF to recover accurate 3D geometery even in persistently shaded areas and also allows shadow-free novel view synthesis [15].

B. Temporal and Seasonal Dynamics

In multi-date photogrammetry, image datasets are often collected over a long period of time, capturing the same scene at different times of the year. This long time span introduces notable appearance differences, especially in the case of seasonal changes such as the vegetation shifting from green to brown in autumn, or the presence of snow during winter.

Earlier Earth observation frameworks, such as S-NeRF and Sat-NeRF, use explicit solar angle modelling to reduce illumination differences, but often struggle in handling seasonal changes. Solar angles are weakly related to the seasonal

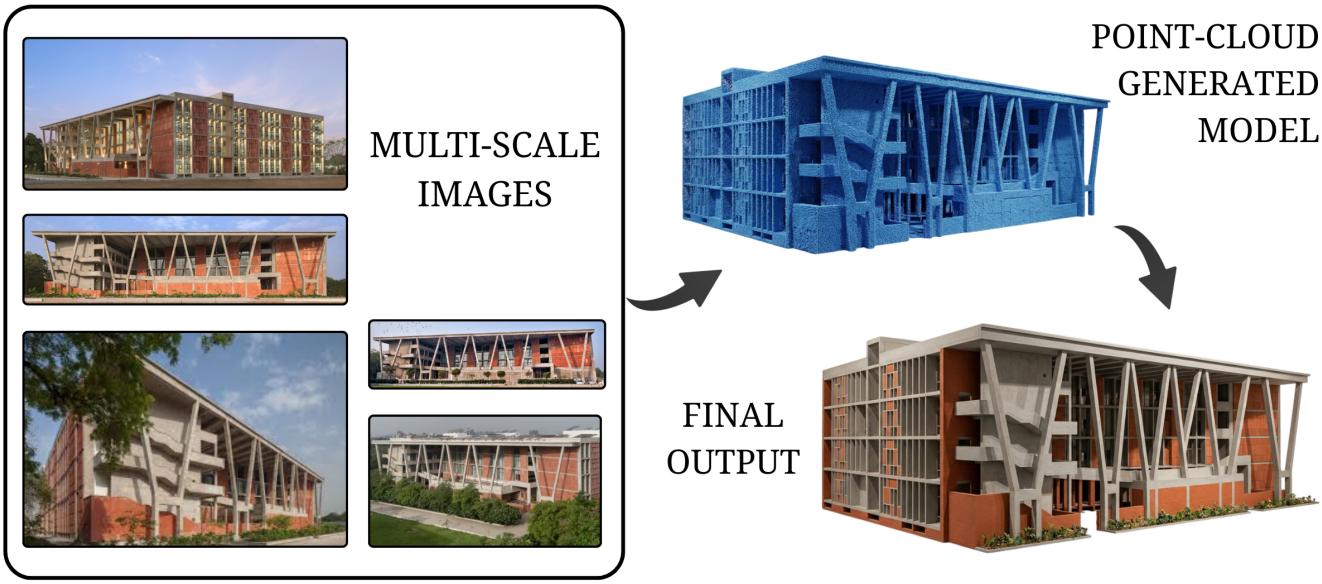

Fig. 7. Overview of traditional 3D reconstruction: Multi-scale 2D images are processed into a sparse or dense point cloud, which is then used to generate a geometric model. Note the limitations in photorealism and heavy computational requirements compared to modern neural rendering techniques.

variations, so the models that rely only on solar position cannot completely correct the pixel consistencies that are caused by the seasonal variations.

To overcome this problem, methods like Season-NeRF are often used, which add time to the rendering framework. This means the model learns to notice things that are specific to certain seasons. This way, the network can put together how things look at times and how the sun is shining from different angles. The Season-NeRF method is really good at combining these things, like how things look at times of the year, and the Season-NeRF method is also good at handling different solar angles, which is important for Season-NeRF. This enables it to stimulate how geographic regions change across different seasons [7].

C. True Digital Orthophoto Map (TDOM) Generation

Traditional Generation of True Digital Orthophotography Maps (TDOMs) mainly depends on Digital Surface Models (DSMs) and complex visibility analysis to handle occlusions. However, this traditional pipeline often produces artifacts like ghosting (disappearance of the subject), geometric distortions, and disappearing edges around building boundaries. To overcome this limitation, frameworks like Ortho-NeRF have been developed, which eliminate the need for explicit geometric priors and traditional occlusion detection [8], [9].

Ortho-NeRF models complex aerial scenes using a continuous implicit neural representation, and also divides large environments into smaller tiles/patches for better efficient processing. To generate distortion-free maps, this framework uses the true-ortho volume rendering technique instead of standard perspective camera projection [9]. Instead of tracing the rays from the central camera viewpoints, this model explicitly casts rays upwards in parallel from normalized ground space. By sampling along these rays through an optimized radiance field, the model inherently resolves complex visibility relationships [8], [9]. This vertical parallel projection removes perspective distortion and helps in reducing the geometric error, holes, and rough building boundaries, which are commonly found in traditional photogrammetry methods.

D. Scaling to Urban Environments

Scaling standard Neural Radiance Fields (NeRFs) to large, city-scale environments is computationally very challenging. These models often struggle to capture the large geometric complexity and fine details of entire cities present within a single network, because of the GPU memory limits and very long training duration.

To overcome these limitations, the researchers have utilized a divide-and-conquer chunking strategy to process large aerial datasets more efficiently. Frameworks like Block-NeRF and Mega-NeRF scale volume rendering to large environments by dividing the scenes into smaller geographical blocks, which can be processed independently [11]. Instead of depending on a single massive network, each of the individual sub-regions is assigned its own NeRF model, which is trained separately [4], [11].

This approach effectively reduces the computational load by distributing, which allows the model to be more optimized in parallel while bypassing the memory bottleneck that is typically seen in large-scale 3D modelling. During the inference, the appropriate network is chosen and rendered based on the location of the camera within the partitioned grid. By distribution of the scene representation across multiple networks these chunking methods enables scalable modeling of large city environments, allowing continuous virtual fly-

throughs and realistic novel view synthesis across entire urban areas. [4], [11]

VI. 3D Gaussian Splatting (3DGS) in Earth Observation

A. Adapting 3DGS for Satellite Imagery

The adaptation of 3D Gaussian Splatting (3DGS) to the satellite domain requires resolving the limitations of the standard pinhole camera model, which is insufficient for the Rational Polynomial Coefficient (RPC) camera models typical of satellite sensors [7], [10]. The EOGS (Earth Observation Gaussian Splatting) framework handles this by replacing resource-heavy RPC projections with approximate affine camera models, mapping 3D points to 2D image coordinates quickly [7]. Guided by this, EOGS++ removes the need for extra preprocessing by combining internal bundle adjustment (BA) directly into the training pipeline [7]. This internal BA increases optical flow estimation to align rendered and reference images, correcting geolocation errors that would otherwise degrade reconstruction quality [7]. Additionally, EOGS++ introduces the **3-PAN** strategy, which reproduces high-resolution single-channel panchromatic (PAN) data into a three-channel representation [7]. This allows the model to directly visualize high-resolution data without needing any external pansharpening of lower-resolution multispectral imagery, which commonly introduces noise [7].

B. Appearance-Adaptive and Shadow 3DGS

Satellite datasets are identified by extreme appearance variations due to their wide temporal span, displaying as seasonal vegetation shifts and dynamic shadows [7]. The SatGS framework addresses these challenges through a hierarchical strategy for appearance-adaptive adjustment [7]. It uses global feature embeddings for each training image to capture large-scale seasonal color shifts, supported by local embeddings for each Gaussian to optimize spatial-specific variations like differential snowmelt [7]. To handle changes in lighting conditions, SatGS incorporates solar angle inputs into a specific multilayer perceptron (MLP) to predict shadow colors [7]. These shadow features are layered on top of seasonal colors to accurately model the shape and hue of shadowed regions [7]. Additionally, SatGS combines uncertainty modeling using features from pre-trained extractors like DINOv2 to identify and ignore temporary objects such as moving cars, ensuring they do not introduce noise into the geometric reconstruction [7].

C. Mitigation of Sparse View Degradation

A difficult challenge for 3DGS in remote sensing is the weakening of geometry in sparse-view scenarios, which normally occurs at the boundaries of flight strips [14]. The Few-Shot Gaussian Splatting (FSGS) method reduces this by introducing a proximity-guided unsampling strategy for completion, allowing new ellipsoids to grow in low-sampled regions [14]. FSGS also implements monocular depth supervision from pseudo-novel views, applying geometric consistency even when multi-view overlap is as low as 2–5 views [14]. The HyGS-TDOM framework uses a hybrid approach, partitioning scenes based on image overlap [14]. It applies standard 3DGS to dense central regions while using the FSGS pipeline for sparse boundary areas [14]. This region-specific optimization allows for highly accurate reconstruction across the entire scene, significantly outperforming standard 3DGS in sparse conditions [14].

D. Gaussian-Based Orthophoto and Mesh Generation

Generating high-fidelity True Digital Orthophoto Maps (TDOMs) and building meshes requires accurate geometric control and parallel processing for large-scale environments [11], [14]. The Ortho-3DGS framework improves 3D building reconstruction by replacing the standard perspective projection with a flat parallel view, which helps in removing building facade irregularities during view synthesis [8]. To further refine geometry, Ortho-3DGS uses depth regulation via pre-trained models like DepthAnything to limit the placement of Gaussian ellipsoids [8]. For large-scale urban reconstruction, the PG-SAG pipeline introduces semantic-aware clustering [11]. It employs models like Language Segment Anything (LSA) to segment building masks and then aggregates these regions into subgroups for parallel optimization [11]. Final surface models are normally generated by visualizing depth maps from these optimized Gaussians and applying Truncated Signed Distance Function (TSDF) fusion to produce smooth, colored building meshes [13]. These advancements allow for rapid, automated 3D modeling of complex urban environments from a simple set of coordinates or building names [13].

VII. Comparative Analysis: NeRF vs. 3DGS

The shift from an implicit continuous function to an explicit point-cloud-based representation showcases a major change in 3D Reconstruction. While the use of a neural network for photorealistic novel view was introduced by Neural Radiance Fields (NeRF), on the other hand, 3D Gaussian Splatting (3DGS) emerged as a strong alternative that overcomes many of NeRF's key limitations.

A. Computational Efficiency

The main difference between 3DGS and NeRF lies in their computational requirements. NeRF represents a scene using a Multilayer Perceptron (MLP), which requires the entire network to be queried a number of times for sample points along each camera ray during volume rendering. This highly dense sampling creates and heavy computational overhead, which often results in several hours of training time, and rendering speed that takes seconds or minutes per frame [9]. For example, on a large-scale urban reconstruction, NeRF required more than 16 hours of training on a single GPU [1].

In contradiction, 3DGS uses an explicit scene representation, which is made up of 3D Gaussian ellipsoids. These ellipsoids are directly projected onto the 2D image plane using a highly parallelized, tile-based rasterizer, which prevents the unnecessary calculation in empty space [8], [9]. This

explicit representation greatly reduces the training time and also enables real-time rendering at speeds of over 100 frames per second (FPS) [7], [14].

B. Reconstruction Quality

Both NeRF and 3DGS models exhibit exceptional capabilities in capturing complex view-dependent effects and lighting. However, 3DGS outperforms NeRF in quantitative image quality metrics—such as PSNR, SSIM, and LPIPS—particularly in large-scale urban environments [1].

- 3DGS outperforms in preserving high-frequency structural features effectively, generating sharper boundaries around building edges, rooftops, and fine surface textures [11]. On the other hand, NeRF models often struggle to represent high-frequency scene features, which may lead to blurred edges, visual artifacts, or overly smooth structures [7], [17].
- The traditional photogrammetry often fails in the areas that do not have any texture, but both of these models (i.e., 3DGS and NeRF) have the capability to recover continuous geometry even in uniform regions such as water bodies or plain walls [3], [8], [9]. However, a well-known limitation of the standard 3DGS is that it can suffer from geometry distortion and excessive Gaussian expansion when only sparse views are available for learning. To achieve competitive accuracy comparable to that of the NeRF models in such cases, 3DGS often requires additional depth supervision or few-shot regularization techniques [8], [14].

C. Editability and Explicit Control

The underlying architecture of these two models affects their post-processing flexibility. NeRF models often behave like a **black-box**, where the features from the scenes are embedded into implicit weights of the MLP. As a result, segmenting and isolating specific objects, editing the scene, or removing noise becomes difficult. Furthermore, combining the large chunks that are split into separate MLP chunks can cause inconsistencies at the boundaries [8].

3DGS provides much greater flexibility for editing and controlling the scene, due to its explicit point-based representation [8]. Individual 3D Gaussians can easily be modified, removed, or classified using spatial bounding boxes or semantic masks [11], [14].

As shown in Table IV, 3DGS consistently outperforms NeRF across all datasets, achieving higher PSNR and SSIM values along with significantly lower LPIPS scores, indicating superior reconstruction quality [1].

TABLE IV
RECONSTRUCTION RESULT COMPARISON BETWEEN NERF AND 3DGS (PSNR IN DB, SSIM AND LPIPS DIMENSIONLESS) [1]

Dataset	Metrics	NeRF	3DGS
Bank of China Tower	PSNR	22.566	**34.1**
	SSIM	0.546	**0.980**
	LPIPS	0.1764	**0.0386**
Gardens by the Bay	PSNR	24.296	**35.6**
	SSIM	0.730	**0.985**
	LPIPS	0.1058	**0.0224**
Tokyo Tower	PSNR	21.094	**32.6**
	SSIM	0.639	**0.968**
	LPIPS	0.1066	**0.0436**

VIII. OPEN CHALLENGES AND FUTURE DIRECTIONS

While neural rendering methods like NeRF and 3DGS have significantly improved the quality of satellite-based 3D reconstruction, several research challenges still exist within this field.

Current methods focus on improving geometric reconstruction and rendering speed, but integrating remote sensing modalities, dynamic scene understanding, and geospatial compatibility remains an open research challenge. Solving these challenges will help to apply neural rendering frameworks to practical Earth observation tasks like digital twin generation, large-scale urban mapping, and geographic analysis.

A. Multi-Model Integration

The current neural rendering methods primarily rely on optical satellite and UAV imagery. While remote sensing platforms provide a vast range of sensing modalities, including LiDAR point clouds, Synthetic Aperture Radar (SAR), and multispectral imagery [1], [5]. Different data sources can help provide important information about complex scene geometry, material reflectance, and absolute elevation [5], [8]. However, integrating these various data types into NeRF or 3DGS frameworks remains a major challenge due to the significant differences in spatial resolution, sensing geometry, and sensor noise.

For example, current Earth observation models commonly struggle to exploit high-resolution panchromatic (PAN) and low-resolution multispectral (MSI) data at the same time (Ma et al., 2025). For avoiding artifacts and inconsistencies caused by traditional pansharpening, many frameworks rely on **Direct Rendering** using PAN data alone [10]. Future directions of research may study hybrid approaches that combine neural rendering with multi-modal sensor fusion to improve geometric accuracy [2], [5], [14]. Such techniques may significantly improve reconstruction accuracy in challenging scenarios like limited optical observations, complex urban occlusions, or unfavourable atmospheric conditions [4], [5], [14]

B. Improved Handling of Transient Elements

Satellite and UAV image collections frequently contain dynamic or temporary elements like moving vehicles and seasonal vegetation changes [4], [7]. Because standard neural rendering frameworks assume a static scene, these transient elements can cause artifacts or inconsistencies during optimization. Recent approaches Sat-NeRF and SatGS, address this issue through uncertainty modelling and visibility filtering that reduce the influence of unreliable pixels [4], [7], [15]. Specifically, SatGS leverages semantic features from a pre-trained DINOv2 extractor to generate visibility maps that allow

the network to effectively ignore regions occupied by transient objects [7]. Additionally, mechanisms like opacity resets in EOG++ help suppress geometrically unstable high-frequency details, like fine vegetation [10]. Future research is expected to focus on integrating explicit temporal modelling and change-detection mechanisms to fully decouple dynamic objects from permanent scene structures [4], [7], [18].

C. Integration with Geographic Information Systems (GIS)

For neural rendering models to be useful in real-world remote sensing applications, they must combine with Geographic Information Systems (GIS), as NeRF and 3DGS frameworks remain isolated with local computer vision coordinate systems [5], [8]. These systems are usually initialized using standard Structure-from-Motion (SfM) modules, but they do not inherently align with geodetic reference frames required for precise mapping and surveying [8]. This gap is filled through the use of advanced georeferencing techniques that contain Rational Polynomial Coefficient camera models, GNSS/GPS data, and Ground Control Points (GCPs) directly into the neural optimization process to achieve absolute geographic precision [8], [10]. Recent techniques have shown that such integration allows for high-precision georeferenced outputs that meet the rigorous standards of the surveying and mapping domain [9]. Furthermore, the application of neural rendering in geospatial workflows depends on the ability to export these continuous representations into standard GIS formats, like high-fidelity point clouds, Digital Surface Models (DSMs), and 3D meshes [2], [5], [13].Pipelines like GBM and Ortho-3dGS highlight how these models generate essential products, including True Digital Orthophoto Maps (TDOMs) and building-scale digital twins. Those are crucial for urban planning, disaster management, and administrative tasks [8], [13], [14]. Standardized export routines will help transition of NeRF and 3DGS from experimental tools to reliable components of global geospatial infrastrucuture [1], [8]

IX. Conclusion

This survey describes recent developments in neural rendering techniques for satellite remote sensing. Particularly, Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS) techniques are considered in this survey.

These techniques represent a shift from traditional photogrammetric reconstruction to learned scene geometry, and appearance is learned directly from multi-view imagery. NeRF provides a method that synthesizes original views with photorealistic quality using implicit volumetric models. Compared to NeRF, 3DGS introduces an explicit Gaussian-based representation that provides faster training and real-time rendering. Based on the literature, these methods solve several major issues in the field of remote sensing, including multi-temporal radiometric inconsistency, top-down bias in vertical reconstruction, processing large-scale urban data, and multi-sensor fusion. Specifically, Gaussian-based frameworks show strong potential for scalable and efficient reconstruction of large environments from satellite images.

Although these improvements have been achieved in the remote sensing area but further research is required to improve the level of multi-modal data integration, handling of transient objects in the scene, and the compatibility of the approach with geospatial technologies like Geographic Information Systems (GIS). In general, the potential of neural rendering methods, specifically those that are based on Gaussian representations, has a lot of potential for the development of next-generation satellite-based 3D reconstruction and digital twin technologies.

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